# An independent reimplementation of Newton et al. (2026): validation status and open questions

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**Date**: 2026-08-05. **Regarding**: Newton, A., Rose, S. C., Kıroğlu, F., Hoang, B.-M., & Rasio, F. A. 2026, *ApJ*, 1006:184, "Intermediate-mass Black Hole Formation from Hierarchical Mergers in Galactic Nuclei" (hereafter N26). **Code**: https://github.com/Menrae/imbh-galactic-nuclei (full history, tests, and a running log of every modeling judgment call in `paper/limitations.md`).

## Why this memo exists

Over the past several weeks I rebuilt N26's model from its equations up, specifically to (a) check whether the paper's central claims reproduce under an independent implementation, and (b) use the validated pipeline to chase down the two open questions N26 poses in Section 5.4. Both of those questions turn out to have answers under this pipeline — and, tellingly, both hinge on the same equation ambiguity below (Section 2.2) — which is a large part of why this seemed worth writing up rather than letting it sit in a private repo. This memo summarizes where validation stands, three places I ran into genuine ambiguity or gaps in the underlying equations that materially affect the results, and what my extensions found. I'd welcome your read on all three, particularly the two ambiguities in Section 2 — in both cases I made a specific choice but neither is fully settled, and you're the people who'd know.

## Questions for you

1. **Eq. 21's exponent** (§2.1): printed as 1/3, but 1/2 in the Volonteri et al. 2013 formula it cites — which was intended?
2. **Eq. 22's** $\langle M_{\rm avg}\rangle/\rho$ (§2.2): star-only, or BH-inclusive? This single choice controls nearly every downstream result below, including both Section 5.4 extensions.
3. **Initial** $a_\bullet/e_\bullet$ **sampling** (§2.3): Section 3 never states this. Is there a convention I'm missing — published or not?

## 1. Validation against Table 1

I ran all four of N26's initial conditions (K20, K20+M, H18, H18+M) at $N = 1000$, $M_\bullet = 4 \times 10^6\, M_\odot$, 10 Gyr, 3 seeds each, and compared against Table 1:

| IC | Mergers (ours, range / Table 1) | Max mass $M_\odot$ (ours, range / Table 1) | % BHs $> 100\,M_\odot$ (ours, range / Table 1) | EMRI % (ours) |
|---|---|---|---|---|
| K20 | 24.0 [14–34] / 34 | 39.1 [30.7–51.4] / 28.4 | 0.0% [0.0–0.0%] / 0% | 29.6% |
| K20+M | 28.7 [19–39] / 30 | 46.2 [38.0–52.2] / 57.8 | 0.0% [0.0–0.0%] / 0% | 31.9% |
| H18 | 1022.0 [923–1102] / 371 | 58,034 [9,181–89,870] / 407.3 | 12.5% [11.7–13.4%] / 7.8% | 34.7% |
| H18+M | 1046.7 [786–1199] / 535 | 11,881 [3,556–16,137] / 526.0 | 12.8% [12.6–13.2%] / 14.3% | 35.0% |

*N26's Table 1 doesn't state whether these are single runs or averages over multiple realizations — I found no mention of seeds, realizations, or repeated runs anywhere in the paper's text, so I'm treating them as single runs and showing my own seed-to-seed range for context, not a like-for-like noise-floor comparison.*

**K20 and K20+M reproduce well** — mergers and max mass both within ~1.3-1.5x of Table 1, which is well inside the plausible range given seed-to-seed variance and the reconstruction uncertainty in the K20 mass distribution itself (N26 cites this as drawn from a simulation (Kremer et al. 2020) rather than a closed-form distribution, so I read the numbers directly off the published histogram).

**H18 and H18+M are a mixed picture.** The population-level statistic that matters most for the paper's headline claim — the fraction of BHs exceeding 100 $M_\odot$ — is actually close (12.5% vs. 7.8% for H18; 12.8% vs. 14.3% for H18+M, the latter within seed noise). But the single "max mass" order statistic is off by 20-220x, and merger count by ~2.9x. I traced this to a specific, still-open mechanism: a genuine, moderately broad heavy tail, not a rare fluke — across the six H18/H18+M seeds, 44-58 of the 1000 BHs exceed 200 $M_\odot$, 13-19 exceed 500 $M_\odot$, and 6-10 exceed 1000 $M_\odot$, via a runaway positive-feedback loop in which both growth channels' efficiency scales with the growing BH's own mass. Only the single most extreme BH per run drives the eye-catching 10,000-90,000 $M_\odot$ figures in the table above, reaching 60-230 successive merger generations along the way (Table 1's own stated maximum is 12-16). I narrowed one contributing factor (a since-fixed bug in my own `mean_bh_mass` placeholder made the runaway tail *more* severe once corrected, implicating the GW-capture channel's own mass-dependence, Eqs. 4-7), but the residual is not fully resolved and I am reporting it candidly as an open item rather than a solved discrepancy.

Full trace: `paper/limitations.md#phase2-emri-rate-high`.

**Bottom line**: bulk-population behavior for both mass families is plausible by the metrics Table 1 reports directly; one bulk statistic Table 1 doesn't report — EMRI fraction — is not: I get 30-37% over 10 Gyr across all four initial conditions (see table above), versus N26's own implied rate of roughly 4-5% over the same span, converting from their ~4-4.8 $\mathrm{Gyr}^{-1}$ merger-rate figures in Section 5.4 (my conversion, not a fraction N26 states directly). The extreme upper tail specifically, for the heavier H18 family, runs hotter than Table 1 in my implementation, for reasons I can point to mechanistically but haven't fully closed the loop on.

## 2. Open questions on the underlying model

The following are cases where my implementation made a specific choice that I believe is defensible from your own cited sources — but I want to flag them rather than assume I read them correctly, since each one turns out to matter a great deal for the results below. The first two are genuine textual ambiguities (I found real support for more than one reading); the third is different in kind — a place N26's text is silent rather than ambiguous.

### 2.1 — Eq. 21's prefactor exponent: 1/3 as printed, vs. 1/2 in the cited source

Eq. 21 (BH spin evolution from a stellar collision) is presented as extending Volonteri, Sikora, Lasota, & Merloni 2013 (ApJ, 775, 94). The published Eq. 21 prefactor is $r_{\mathrm{ISCO}}^{1/3}/3$ (I confirmed this isn't an OCR artifact via a high-resolution crop of the PDF). Volonteri et al. 2013's own Eq. 14 — the Bardeen (1970) thin-disk spin-up result — has exponent 1/2: $r_{\mathrm{ISCO}}(t)^{1/2}/3$. The inner square-root term is identical between the two papers once N26's unsubscripted "$r$" is read as $r_{\mathrm{ISCO}}$, so this looks like a single exponent that doesn't match, not a different formula.

I implemented 1/2, since it's the long-established Bardeen result used consistently across the literature (King & Kolb 1999; King, Pringle & Hofmann 2008; Volonteri et al. 2005, 2013), and Eq. 21's text explicitly attributes the formula to Volonteri et al. 2013 rather than deriving a new variant. My best guess is that this is a typesetting slip rather than an intentional change — but I haven't confirmed that with you, and it's exactly the kind of thing that's very easy to miss without going back to Eq. 14 of the cited source directly. Could you confirm which exponent was intended? It has a direct, first-order effect on predicted maximum spins.

### 2.2 — Eq. 22's $\langle M_{\mathrm{avg}} \rangle$ and $\rho$: star-only, or BH-inclusive?

This is the more consequential of the two ambiguities. Eq. 22's two-body relaxation timescale, $t_{\mathrm{relax}} = 0.34\,\sigma^3/(G^2\rho\langle M_{\mathrm{avg}}\rangle \ln\Lambda)$, depends on an "average object mass" $\langle M_{\rm avg}\rangle$ and density $\rho$ that N26's text describes only generically ("the average object mass," "their mass density"). I found textual support for two different readings, and — as far as I can tell — no way to settle it from the paper's text alone:

- **Star-only** ($\langle M_{\rm avg}\rangle = 1\,M_\odot$, $\rho = \rho_\star$): S. C. Rose et al. 2022 (ApJL, 929, L22) — the paper N26 explicitly says this relaxation mechanism is "first developed by" — states its own version of this formula (their Eq. 10) is explicitly for "a **single-mass system**," with $\langle M_*\rangle$ "here assumed to be 1 $M_\odot$." Rose et al. 2022 has no BH-density term at all, so its formula can't mean anything else.
- **BH-inclusive** (density-weighted mean over both stars and background BHs, using N26's own Eq. 2 BH-density profile): N26's Eq. 22 text drops Rose et al. 2022's "single-mass system" qualifier, and N26 introduces a genuinely new two-component density apparatus that Rose et al. 2022 never had. Most tellingly, Eq. 23 (mass-segregation time) explicitly writes $t_{\rm seg} \approx (M_\star/m_{\rm BH}) \times t_{\rm relax}(\langle M_{\rm avg}\rangle = M_\star, \rho = \rho_\star)$ — an explicit override to star-only for that one derived quantity, which only makes sense as a deliberate exception if Eq. 22's own default isn't already star-only.

I tested both readings empirically (N=1000, H18, 10 Gyr): star-only gives the Section 1 H18 numbers above (34.7% EMRI, 1022 mergers); BH-inclusive gives ~66% EMRI (64.8-67.5% across 3 seeds) and only 1-2 mergers per run, reproducing what looks like the exact early-development problem this relaxation mechanism was meant to fix — objects get diffused into the SMBH before the merger channel can ever complete more than a handful of events. Neither reading reproduces Table 1's exact balance, but only star-only leaves the merger channel functioning at meaningful scale, so it's my working default. This is logged as an open, unresolved choice, not a settled one (`paper/limitations.md#average-object-mass`).

I'd be glad to know which reading was intended, or whether Eq. 22 was meant more loosely than either extreme — because, as it turns out, this single choice controls nearly every downstream result I describe below.

### 2.3 — Initial orbital properties: a gap, not a discrepancy

Unlike 2.1 and 2.2, this isn't a case where N26 says something and I read it a particular way — it's a place the text is silent. Section 2 states BHs' initial masses, spins, and orbital properties are drawn "statistically as described in Section 3," but Section 3, read in full, covers only mass and spin distributions (K20, H18, the primordial-binary fraction); it never states how the initial semimajor axis $a_\bullet$ or eccentricity $e_\bullet$ about the SMBH are sampled. I confirmed this directly against the PDF rather than assuming it from a paraphrase.

Since N26 explicitly presents its dynamical framework as an extension of Rose et al. (2022)'s model, I inherited that paper's stated convention: thermal eccentricity, and semimajor axis log-uniform out to 0.1 pc (N26's own stated focus region for the outer bound). Neither paper states an inner bound; I picked $a_{\rm min} = 10^{-3}$ pc from my own inspiral-time argument (a BH born there takes far longer than 10 Gyr to inspiral via quiescent GW decay alone, so it doesn't manufacture EMRIs as a numerical artifact) — but that choice is mine, not N26's or Rose et al.'s, and it directly sets the EMRI rate I get. If there's a value or derivation I'm missing — from either paper, or unpublished — I'd want to use it instead; it's the most likely single lever to close the EMRI-rate gap in Section 1's bottom line.

## 3. Extending the validated model to N26's own Section 5.4 open questions

With validation in hand, I used the same pipeline to pursue the two questions Section 5.4 poses as open future work. Both answers below are reported under **both** readings from Section 2.2, since the choice turned out to matter as much here as it did for validation itself.

### 3.1 — Is there a sharp critical initial-mass threshold, or a gradual one?

Scanning a log-uniform mass family on $[6, m_{\max}]\, M_\odot$ over $m_{\max} = 16$–$100\, M_\odot$: a first pass at 3 seeds/point (matching the Section 1 validation convention) made the transition look sharp, with $m_{\max} = 31.8\, M_\odot$ appearing fully saturated (3/3 seeds producing an IMBH). That turned out to be a small-sample illusion — rerunning the same point at 8 seeds showed it was actually 6/8 (75%), not saturated at all. I'm surfacing this correction explicitly because it's exactly the kind of thing a thin sample can manufacture, and it's why I adopted an 8-seed floor for every location claim from that point on, including the SMBH-mass scan in 3.2 below.

With that floor applied: under the **star-only** reading, the probability that *any* BH out of 1000 crosses 100 $M_\odot$ in a 10 Gyr trial rises smoothly from 0% to 75% across roughly $m_{\max} \approx$ 20-32 $M_\odot$ (mean population mass $\approx$ 12-15 $M_\odot$), without ever fully saturating within that band — a genuine, gradual crossover, not a step function, and one that sits well below H18's own 100 $M_\odot$ upper limit. Under the **BH-inclusive** reading, no threshold appears anywhere in the tested range — even at $m_{\max} = 100$ (H18 exactly), only a marginal 0.1-0.3% of the population crosses 100 $M_\odot$. I also checked N26's own 0%/15% primordial-binary-merger axis (the "+M" prescription) at the same resolution and found no statistically distinguishable effect on the crossover (pooled 19/40 vs. 20/40 any-IMBH trials, Fisher's exact $p = 1.0$) — the mass-scale of the initial distribution, not the primordial-pairing fraction, is what drives it.

### 3.2 — Does the result generalize beyond the Milky Way's SMBH mass?

Scanning $m_{\text{smbh}}$ over 3 decades (7 points, $1.26 \times 10^5$ to $1.26 \times 10^8\, M_\odot$, 8 seeds each, H18, cluster structure held fixed at the Milky Way's): under **star-only**, all 56/56 runs across every mass tested produced at least one IMBH — the paper's qualitative result does not appear to be a Milky Way-specific coincidence. Under **BH-inclusive**, the mean fraction of BHs exceeding $100\,M_\odot$ stayed below 0.5% at every single grid point across the full 3-decade range — the suppression effect generalizes just as robustly as the formation effect does. So the generalization answer is a qualified "yes": the paper's claim holds up across SMBH mass, but which side of the claim you get (forms readily vs. essentially doesn't form) still depends on the Section 2.2 ambiguity.

One further, more tightly-scoped observation from this scan: holding cluster structure fixed while varying only $m_{\mathrm{smbh}}$ (a modeling choice on my end, not from N26) revealed a non-monotonic runaway-growth regime concentrated 3-10x *below* the Milky Way's own SMBH mass, where individual BHs reached masses of millions of $M_\odot$ — in some runs, exceeding the central SMBH itself. Eleven of the 112 runs in this scan (all star-only, all in the $4 \times 10^5$-$1.26 \times 10^6\,M_\odot$ band where this effect concentrates) didn't reach 10 Gyr before hitting the integrator's step ceiling — their reported masses reflect partial integration (roughly 1-9 Gyr), not the full window. I checked this wasn't manufacturing the finding: a seed that completed *within* the cap at the same mass still reached $1.56 \times 10^6\,M_\odot$, and the ceiling-hit rate itself is non-monotonic (0% at the lowest mass tested, 50-87.5% in the band, back to 0% at and above the Milky Way anchor) — not the shape you'd expect if this were simply numerical stalling getting worse as the calculation gets harder. That's the basis for my confidence that it isn't purely an integration artifact, though the partial integration does mean the true (uncapped) masses in this band are a lower bound, not a final answer. I'm flagging the whole finding as a consequence of holding the cluster's density profile fixed while shrinking the SMBH — not a claim about what a real lower-mass galactic nucleus would do — and mention it mainly because it touches the same growth channels as the Section 1 validation residual.

## Closing

I'd welcome any correction, context, or unpublished detail that bears on any of the three open items in Section 2 — particularly if one was a known issue, if there's a reading of Eq. 22 I haven't considered, or a convention for initial orbital sampling I've missed. The full repository (code, tests, every raw run, and `paper/limitations.md`'s complete log of every judgment call made along the way) is public and linked above if useful. Happy to share more detail on any of the above, or to run additional checks if something here looks off.